

Lecture 17 – Bending – problem solving tutorial

1. Draw FBD
2. Write down all static equilibrium equation(s)
3. Calculate degree of static indeterminacy
4. Find out the compatibility equation(s)

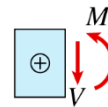
$$\delta = 0, \quad \delta_{A,\text{left}} = \delta_{A,\text{right}}, \quad F_A = k\delta_A$$

5. Using Force-displacement relation, convert compatibility equation(s) into equations of reaction force(s)

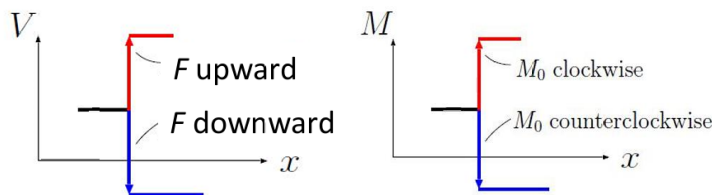
$$\delta = \frac{5qL^4}{384EI}, \quad \delta = \frac{qL^4}{8EI}$$

6. Solve all reaction force(s)
7. Draw SFD and BMD

$$\frac{dV}{dx} = -q(x), \quad \frac{dM}{dx} = V(x)$$



- a) Method of jumping, from left to right



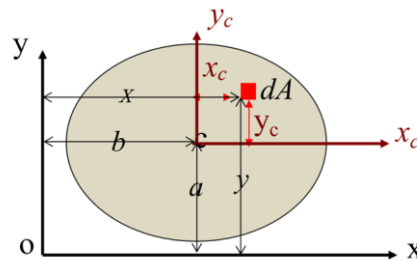
- b) Method of sections

8. Calculating moment of inertia

$$I = \frac{bh^3}{12}, \quad I = \frac{\pi r^4}{4}$$

Centroid:

$$\left. \begin{aligned} x_C &= \frac{A_1 x_{1C} + A_2 x_{2C} + \dots + A_n x_{nC}}{A_1 + A_2 + \dots + A_n} \\ y_C &= \frac{A_1 y_{1C} + A_2 y_{2C} + \dots + A_n y_{nC}}{A_1 + A_2 + \dots + A_n} \\ z_C &= \frac{A_1 z_{1C} + A_2 z_{2C} + \dots + A_n z_{nC}}{A_1 + A_2 + \dots + A_n} \end{aligned} \right\} \begin{array}{l} \text{2D/plane} \\ \text{figure} \end{array}$$



The parallel axis theorem: $I_X = I_{X_C} + Ad^2$, I_{X_C} : moment of inertia of X_C axis parallel to X axis passing its centroid; A : total area; d : distance between two parallel axis X and X_C .

9. Calculating stress and strains from V and M

$$\sigma_x = -\frac{My}{I}, \quad \epsilon_x = -\frac{My}{EI}$$

$$\tau_{\max} = \frac{3}{2} \tau_{\text{avg}}$$

10. Calculating the deflection and deflection angle

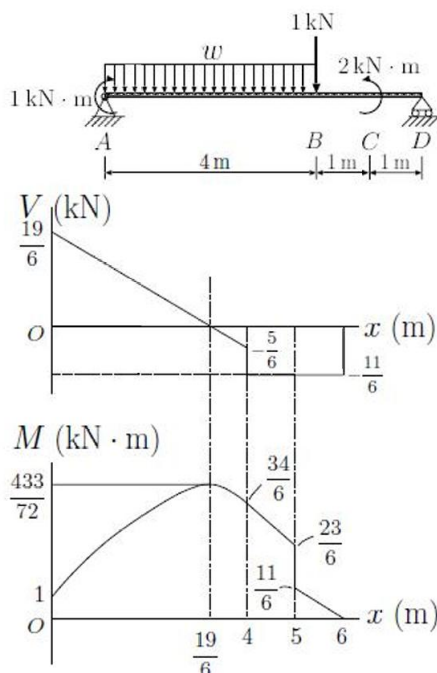
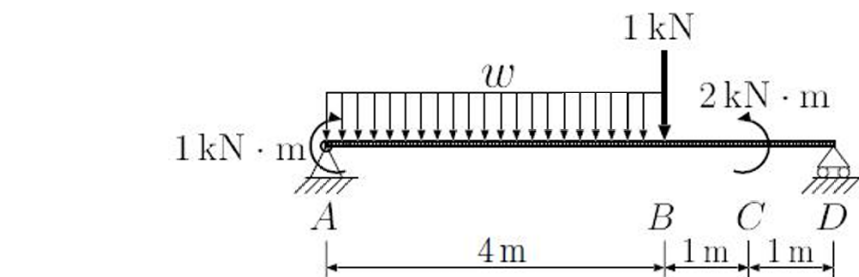
a) Method of successive integrations

$$EI \frac{d^2 v}{dx^2} = M \Rightarrow EI v = \int_0^x \int_0^x M(x) dx dx + C_1 x + C_2$$

b) Method of superpositions

Practice:

Shown in figure below is a simple beam with various loadings applied. The distributed force is applied on segment AB with a constant intensity of $w = 1 \text{ kN/m}$. Draw the shear and bending moment diagrams, and on the plots mark the characteristic values, i.e., values at *transitions and boundaries*, *maximum absolute values* and their corresponding *locations*.



Solution:

1) Equilibrium equation

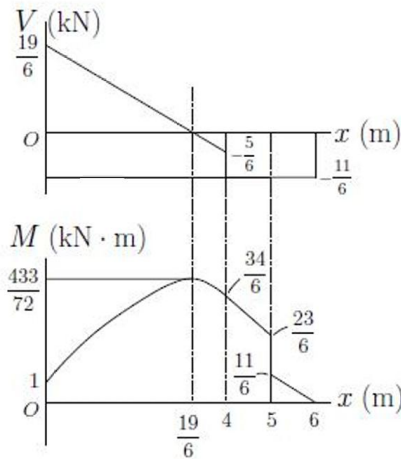
$$M_A = 0 \rightarrow R_D = 11/6 \text{ kN}$$

$$R_A + R_D = 5 \rightarrow R_A = 19/6 \text{ kN}$$

2) V and M diagrams by differential relations and jumps rules

3) Or by method of sections, V and M in a piece-wise form

4) the piece-wise form of V and M



$$V(x) = \begin{cases} (\frac{19}{6} - x) \text{ kN}, & 0 \leq x < 4 \\ -\frac{11}{6} \text{ kN}, & 4 < x \leq 6 \end{cases}$$

$$M(x) = \begin{cases} (\frac{19}{6}x - \frac{x^2}{2} + 1) \text{ kN} \cdot \text{m}, & 0 \leq x \leq 4 \\ (13 - \frac{11}{6}x) \text{ kN} \cdot \text{m}, & 4 \leq x < 5 \\ (11 - \frac{11}{6}x) \text{ kN} \cdot \text{m}, & 5 < x \leq 6 \end{cases}$$

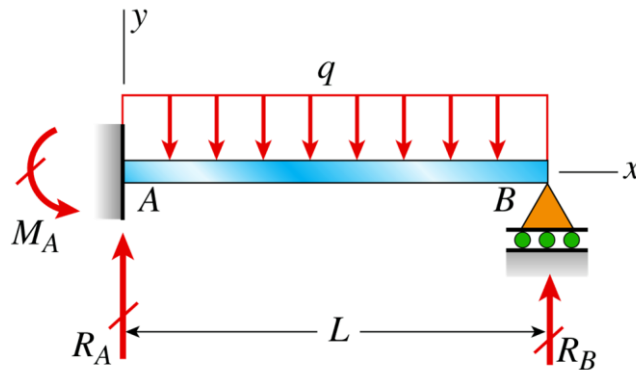
5) The maximum stress, if we know the cross-sectional information

$$\sigma_m = -\frac{My}{I}$$

$$\tau_m = \frac{3}{2} \frac{V_m}{A}$$

2. A propped cantilever beam AB of a length L supports a uniform load of intensity q .

Determine the reactions, shear forces, bending moments, slopes, and deflections of the beam.



Method 1: Analyze this beam by solving the second-order differential equation of the deflection curve.

Equilibrium equation:

$$R_A = qL - R_B, \quad M_A = \frac{qL^2}{2} - R_B L$$

Compatibility equation:

$$\delta_B = v(L) = 0$$

Bending moment:

$$M(x) = R_A x - M_A - \frac{qx^2}{2} = qLx - R_B x - \frac{qL^2}{2} + R_B L - \frac{qx^2}{2}$$

Moment-deflection equation:

$$\begin{aligned}
 EIv'' &= M = qLx - R_Bx - \frac{qL^2}{2} + R_BL - \frac{qx^2}{2} \\
 \Rightarrow EIv' &= \frac{qLx^2}{2} - \frac{R_Bx^2}{2} - \frac{qL^2x}{2} + R_BLx - \frac{qx^3}{6} + C_1 \\
 \Rightarrow EIv &= \frac{qLx^3}{6} - \frac{R_Bx^3}{6} - \frac{qL^2x^2}{4} + \frac{R_BLx^2}{2} - \frac{qx^4}{24} + C_1x + C_2
 \end{aligned}$$

Boundary conditions

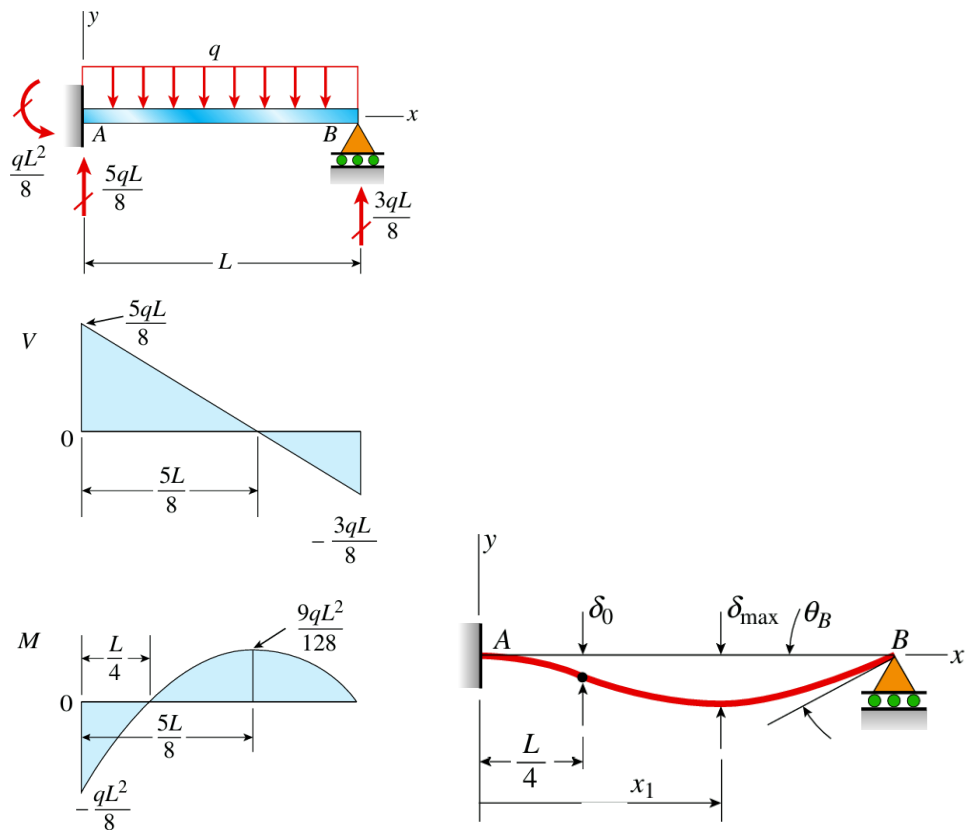
$$\begin{aligned}
 v(0) &= 0, \quad v'(0) = 0 \\
 \Rightarrow C_1 &= C_2 = 0
 \end{aligned}$$

Compatibility equation:

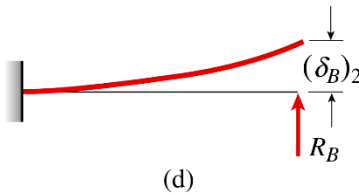
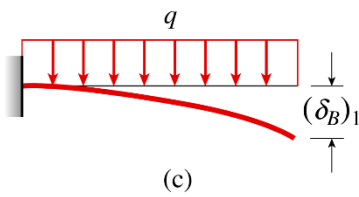
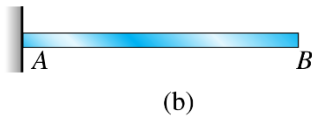
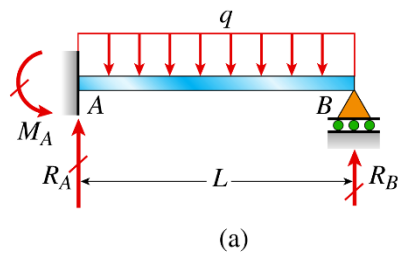
$$\begin{aligned}
 EIv(L) &= \frac{qL^4}{6} - \frac{R_BL^3}{6} - \frac{qL^4}{4} + \frac{R_BL^3}{2} - \frac{qL^4}{24} = \frac{R_BL^3}{2} - \frac{3qL^4}{16} = 0 \\
 \Rightarrow R_B &= \frac{3qL}{8}, \quad \Rightarrow R_A = \frac{5qL}{8}, \quad M_A = \frac{qL^2}{8}
 \end{aligned}$$

Then

$$V = \frac{5qL}{8} - qx, \quad M = \frac{5qLx}{8} - \frac{qL^2}{8} - \frac{qx^2}{2}$$



Method 2: Method of superposition



Compatibility equation:

$$\delta_B = \delta_{B1} - \delta_{B2} = 0$$

From H.1.1 and H.1.4

$$\delta_{B1} = \frac{qL^4}{8EI}, \quad \delta_{B2} = \frac{R_B L^3}{3EI}$$

Then

$$\delta_B = \frac{qL^4}{8EI} - \frac{R_B L^3}{3EI} = 0 \Rightarrow R_B = \frac{3qL}{8}$$

We have

$$v_1 = -\frac{qx^2}{24EI}(6L^2 - 4Lx + x^2)$$

$$v_2 = \frac{R_B x^2}{6EI}(3L - x)$$

Then

$$v = v_1 + v_2 = -\frac{qx^2}{48EI}(3L^2 - 5Lx + 2x^2)$$